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Coupling characteristic analysis and synchronization stability control for Multi-Paralleled VSCs system under symmetric faults

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ABSTRACT

The grid-following voltage source converters (GFL-VSCs) are prone to lose synchronization with the grid under symmetrical faults, which seriously threatens the security of the power system. Different from the existing works, this article takes a multi-paralleled GFL-VSCs system as the research object and deeply explores the power flow coupling mechanism between VSCs and synchronization stability control method for the system during low voltage ride through (LVRT). Firstly, by a nonlinear dynamic model of PLL, the synchronization instability issue of the multi-paralleled system is revealed. Furthermore, the impact of power flow coupling effect on the VSC's synchronization stability is analyzed from the two aspects: the existence of equilibrium points and the transient synchronization behavior. Subsequently, a stability criterion based on coherence properties is constructed, which can quantitatively evaluate the instability risk of the multi-paralleled system. Finally, a current optimization control strategy is proposed, which can ensure the maximum transient stability margin of the entire system and meet the reactive power requirements of the grid code. The simulation results validate the correctness of the theoretical analysis and the effectiveness of the proposed strategy.

1. Introduction

With the development of economy and society, the global demand for energy continues to grow. In order to deal with the severe issues such as of energy shortage and environmental pollution, a large number of renewable energy generation units represented by photovoltaic and wind power are being connected to the grid through voltage-source converters (VSCs) in recent years, which would lead to complex and profound variations in the synchronization stability mechanism of the power system [1]. Unlike the synchronous generators (SGs) that are synchronized with the grid through the actual physical rotor, VSCs often use phase-locked loop (PLL) to achieve grid-connected synchronization [2]. However, due to grid voltage sag [3] and output characteristic change [4], it is not easy for PLL to realize the timely resynchronization, which makes the safe operation of the power system face severe challenges.

At present, the synchronization stability issue of GFL-VSC under grid faults has attracted extensive attention from academia and industry. [5] mentioned the dynamics of the current references can deterioratethe GFL-VSC's small-signal stability. [6] indicated the interaction of PLL and the DC link voltage control loop would influence the dynamic characteristics of GFL-VSC. In [7], it is found that the interaction between PLL and current loop would lead to insufficient damping, resulting in small-signal instability. In addition to the small-signal stability, there are also some studies on transient synchronization stability. [8] found that too deep voltage drop or inappropriate current reference values would cause the system to loses its equilibrium points, resulting in the synchronization instability of VSC. Then, focusing on the VSC's transient synchronization process, [9–13] pointed out that even if the existence of equilibrium points are met, the system cannot guarantee a successful transition to the stable equilibrium point. [9] pointed out that too long controller delay would cause the GFL-VSC to face greater risk of synchronization instability. [10] found that improper PLL integration coefficient would reduce the system damping ratio, which is not conducive to the transient stability of VSC. Based on the phase portraits method, [11] indicated that reducing the settling time would also deteriorate the transient characteristics of the PLL. Additionally, referring to the rotor motion equation of SG and the equal area criterion (EAC), [12] mentioned that if the equivalent potential energy of PLL is not enough to offset the increment of its equivalent kinetic energy, the loss of synchronization (LOS) of VSC cannot be avoided. Furthermore, a stability

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Nomenclature

$\boldsymbol{U_g}$, U_g, θ_g	Vectors, amplitude, and phase angle of the grid voltage.
$\boldsymbol{U_t}$, $\boldsymbol{I}$	Vectors of the terminal voltage and the output current for the VSC.
$\boldsymbol{Z_b}$, R_b, X_b, L_b	Impedance, resistance, reactance, and inductance of the branch line.
$\boldsymbol{Z_p}$, R_p, X_p, L_p	Impedance, resistance, reactance, and inductance of the public transmission line.
θ_{Zb}, θ_{Zp}	Impedance angle of the branch line and public transmission line in the PLL reference frame.
θ_I, θ_{PLL}	Current vector angle and PLL's output angle in the PLL reference frame.
ω_{PLL}, ω_g, ω_b	PLL's output angular frequency, grid voltage vector's angular frequency, and angular frequency base value.
k_p, k_i	PLL's PI coefficients.
δ	Phase angle difference between PLL's d-axis and $\boldsymbol{U_g}$.
C_f, L_f	Filter capacitor and inductance.
P^*_{vi}, P_{vi}	Equivalent prime mover power and equivalent output power of the VSC.
J, D	Equivalent inertia and equivalent damping of the VSC.

Superscripts

*	Reference values

Subscripts

d/q	d/q-axis components in PLL reference frame.
1,2…*i*…*n*	1,2…*i*…*n*-th VSC.

criterion based on the Lyapunov function was constructed in [13], and it was reported that when the initial status is not within the range of attraction region, the GFL-VSC would face the risk of LOS as well. However, all of the above works take a single-machine system as the research object, and the conclusions are not completely applicable to the multi-machines system. Existing exploration work on the synchronization stability of the multi-paralleled VSC system is still in the initial stage, and related research results are relatively few. In [14], based on the Newton-Raphson method, an assessment criterion for the existence of equilibrium points applied to multi-parallel system was established. By comparing the relationship between the impedance voltage drop of the farthest-end converter and the power grid voltage, a synchronous stability level index for the existence of equilibrium points evaluation was proposed in [15]. But the instability criterion for the transient synchronization process was not mentioned, and the explanation of the coupling mechanism is not clear enough. Then, in [16], the EAC was applied to the transient stability evaluation of the multi-VSCs system. However, due to the power-angle coupling in the multi-VSC system, the acceleration and deceleration area of each VSC can't be accurately obtained by integral calculation, and it is difficult to obtain quantitative analysis results, which makes the applicability of this method questionable.

Furthermore, to improve the synchronization stability of the GFL-VSC system, some control schemes have been proposed in the present literatures. In [17], a current control method based on the resistance-inductance ratio was proposed, but it is only suitable for the single-VSC system. [10] adopted an additional damping control strategy, but it could not cope with the instability issue caused by the loss of equilibrium point. An adaptive current algorithm [8] and adaptive power controller [12] are designed, but if the controller parameters are not selected properly, the control performance is difficult to be guaranteed. Some studies mentioned that the synchronization stability of the VSC can be improved by blocking the PLL [18], freezing the PLL integration [19], and adjusting the PLL input [20], respectively, but these methods would change the phase reference of the system, resulting in the actual output current may not meet the reactive power requirements of the grid code. In [14], an autonomic current distribution scheme applicable to the multi-machine system is proposed, but it did not achieve the control objective of minimizing transient instability risk. In addition, [21] also designed a current control method for the converters in the paralleled system, but it may not fully meet the reactive power requirements of the grid code. Consequently, more in-depth research on the stabilization and control for the multi-VSC system is required to meet the rapid development demands of the power system.

To this end, this article aims to further study the synchronization stability of multi-parallel GFL-VSC system, deeply reveal the power flow coupling mechanism, and put forward the transient stability control method. The main contributions of this paper are presented as follows:

1) By the dynamic equation of PLL and equal area criterion, the influence of power flow coupling effect on the existence of equilibrium point and transient synchronization behavior for VSC is revealed.
2) Based on the coherence properties, a reduced-order aggregation model of a multi-VSC system is deduced, and then the transient synchronization instability criterion of the aggregated converter is constructed.
3) To minimize the transient instability risk of the multi-paralleled VSCs system, a current optimization control strategy is proposed, which can significantly enhance the transient synchronization stability of the system during low voltage ride through (LVRT).

The rest of this paper is organized as follows: In section 2, the nonlinear dynamic model of a multi-paralleled VSCs system is built. In section 3, the synchronization stability and power flow coupling effect of the system are studied. In section 4, a current optimization control strategy for the multi-paralleled system is presented. In section 5, the simulation verifications are performed to validate the theoretical analysis and proposed method. In section 6, the conclusion of this paper is given.

2. Modeling of multi-paralleled GFL-VSCs system

2.1. System Description

Fig. 1 shows the structural topology of a multi-parallel GFL-VSCs system. The VSC pass through the LC filter, and then connected to the grid via line impedance. The control system of the VSC is mainly composed of phase-locked loop (PLL) and current loop during LVRT, as shown in Fig. 2. In the pre-fault stage, the current reference value I^*_{dq} is given by power control loops (current reference Flag = 1), and the reactive power Q^* is set to 0 to realize unit power factor control. However, when a symmetric grid fault occurs, the VSC would immediately enter the LVRT stage, and a single current loop control mode (current reference Flag = 2) would be adopted to meet the fast reactive

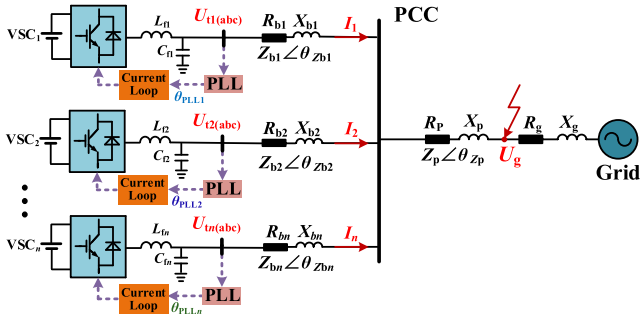


Fig. 1. Structural topology of multi-paralleled GFL-VSC system.

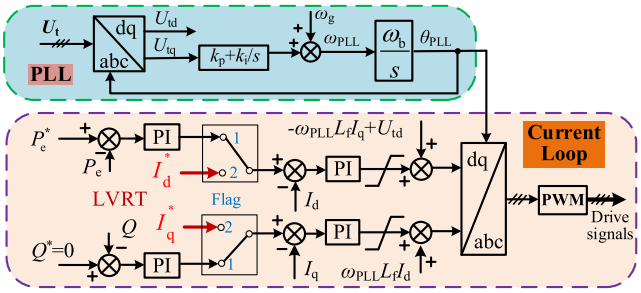


Fig. 2. Control system of the VSC.

current injection requirement of grid code. Hence, the reactive current reference value I_q^* would be designated by the specifications of different grid code [22], and the active current reference value I_d^* is often determined by the residual current capacity.

According to [23], the control system of VSC has multi-timescales characteristics. During LVRT, due to the power loop is out of operation, the PLL timescale and the current loop timescale are mainly considered. Moreover, since the bandwidth of the PLL is much smaller than that of the current loop, it can be assumed that the current loop is ideal when focusing on the synchronization stability issue when taking the PLL as the key factor, that is, the output current value is equal to the current reference value ($I_d = I_d^*$, $I_q = I_q^*$) [14]. In this case, VSC can be further equivalent to a controlled current source during the fault.

2.2. Voltage vector equation

As shown in Fig. 1, each VSC is aggregated to the point of common coupling (PCC) through branch transmission lines, and then connected to the grid via a public transmission line. Taking the i-th VSC as an example, according to the kirchhoff voltage law (KVL), the terminal voltage of the i-th VSC can be obtained,

$$\boldsymbol{U}_{ti} = \boldsymbol{Z}_{bi}\boldsymbol{I}_i^* + \boldsymbol{Z}_p\sum_{j=1}^{n}\boldsymbol{I}_j^* + \boldsymbol{U}_g \tag{1}$$

where $\boldsymbol{Z_{bi}}$, $\boldsymbol{Z_p}$, $\boldsymbol{I}_i^*$ and $\boldsymbol{I}_j^*$ can be further expressed as

$$\begin{cases} \boldsymbol{Z}_{bi} = R_{bi} + jX_{bi} = \boldsymbol{Z}_{bi}\angle\theta_{Zbi}, \boldsymbol{Z}_p = R_p + jX_p = \boldsymbol{Z}\angle\theta_{Zp} \\ \boldsymbol{I}_i^* = I_{di}^* + jI_{qi}^* = \boldsymbol{I}_i^*\angle\theta_{Ii}, \boldsymbol{I}_j^* = I_{dj}^* + jI_{qj}^* = \boldsymbol{I}_j^*\angle\theta_{Ij} \end{cases} \tag{2}$$

2.3. PLL equation

For grid-following VSC, PLL is the key link to realize grid-connected synchronization. There are n GFL-VSCs in this multi-paralleled system, so there are n different PLL dq-axis reference frames, as shown in Fig. 3 (a). Further, taking $\boldsymbol{U_g}$ as the reference vector, define the direction of $\boldsymbol{U_g}$ as the D-axis, which also serves as the unified reference frame of the entire system. Then combined with Fig. 2 and Fig. 3 (a), the PLL equation of the VSC can be given as,

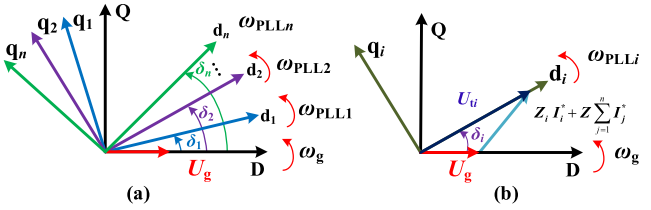


Fig. 3. (a) PLL dq-axis reference frame of each VSC in multi-paralleled system; (b) vector diagram of the the i-th VSC in the PLL dq-axis reference frame.

$$\begin{cases} \dfrac{d\omega_{PLL}}{dt} = k_p\dfrac{dU_{tq}}{dt} + k_iU_{tq} \\ \dfrac{d\delta}{dt} = \omega_b(\omega_{PLL} - \omega_g) \end{cases} \tag{3}$$

where $\delta = \theta_{PLL}$-θ_g, since $\boldsymbol{U_g}$ is coincident with D-axis, θ_g is equal to 0. Moreover, it can be seen from Fig. 3 (a) that the DQ-axis reference frame rotates at the angular frequency ω_g, the dq-axis reference frame rotates at the angular frequency ω_{PLL}. In the steady status, $\omega_{PLL} = \omega_g$, so δ_i is a constant. But in a dynamic process, as $\omega_{PLL} \neq \omega_g$, δ would vary. If $\omega_{PLL} > \omega_g$, δ would continually increase; conversely, if $\omega_{PLL} < \omega_g$, δ would gradually decrease. Consequently, δ could present the synchronization status of the VSC, which can be defined as the VSC's equivalent power angle (EPA).

2.4. Nonlinear dynamic model of the GFL-VSC in Multi-Paralleled system

Fig. 3 (b) shows the vector diagram of the i-th VSC in the PLL reference frame. Based on the principle of coordinate transformation, (1) can be decomposed into dq-axis components, the q-axis component of terminal voltage U_{tqi} can be written as,

$$\begin{aligned} U_{tqi} &= -U_g\sin\delta_i + \underbrace{I_i^*[Z_{bi}\sin(\theta_{Zbi} + \theta_{Ii}) + Z_p\sin(\theta_{Zp} + \theta_{Ii})]}_{U_{tqii}} \\ &\quad + \sum_{j=1,j\neq i}^{n}\underbrace{Z_pI_j^*\sin(\theta_{Zp} + \theta_{Ij} + \delta_j - \delta_i)}_{U_{tqji}} \end{aligned} \tag{4}$$

where $-U_g\sin\delta_i$ represents the impact of the grid voltage status, and can be defined as a grid-synchronization term. U_{tqii} reflects the interaction between the output current of the i-th VSC itself and line impedance, which can be defined as a self-synchronization term. While U_{tqji} characterizes the mutual effect between VSCs, it can be defined as a coupling-synchronization term.

Then, substitute (4) into (3) and simplify, the nonlinear dynamic model of the VSC in multi-paralleled system can be obtained.

$$\begin{cases} J_i\dfrac{d\omega_{PLLi}}{dt} = P_{vi}^* - P_{vi} - D_i(\omega_{PLLi} - \omega_g) \\ P_{vi}^* = U_{tqii} + \sum_{j=1,j\neq i}^{n}U_{tqji}, P_{vi} = U_g\sin\delta_i \\ J_i = \dfrac{(1 - k_{pi}L_{bi}I_{di}^*)}{k_{ii}}, D_i = \dfrac{k_{pi}U_g\omega_b\cos\delta_i}{k_{ii}} \end{cases} \tag{5}$$

It can be seen from (5) that the GFL-VSC in system also has the same form as the rotor motion equation of the SG. Hence, there is a similarity between VSC and SG in the mathematical model when depicting synchronization characteristics. The SG's synchronization stability is affected by the unbalanced active power. Similarly, the equivalent power deviation P_{vi}^*-P_{vi} also has such an impact on VSC's synchronization stability. And the detailed analysis would be carried out in Section 3.

It is worth mentioning that P_{vi}^* and P_{vi} in (5) are not active power in real physical sense, but q-axis components of different voltage vectors in PLL reference frame. P_{vi}^* is the q-axis component of line impedance voltage drop, reflecting the influence of the interaction between output current and line impedance on the transient synchronization characteristics of GFL-VSC. While P_{vi} is the q-axis component of grid voltage, which directly reflects the influence of grid status on the GFL-VSC's transient synchronization stability.

3. Synchronization stability and coupling characteristic analysis

To achieve the synchronization stability of the multi-paralleled system during LVRT, each GFL-VSC is required to have a stable equilibrium

point and a sufficiently good transient synchronization behavior. Moreover, in such a multi-machine system, there are complex coupling effects between VSCs, which would influence the synchronization characteristics of each VSC. Therefore, this section would reveal in detail the impact of power flow coupling on the VSC's synchronization stability.

3.1. Existence of equilibrium points

Regardless of the single VSC or multi-VSCs system, the existence of equilibrium points is the prerequisite for the system to realize synchronization stability during LVRT. According to (4) and (5), if $\boldsymbol{I}_j^* = 0$ (j = 1,2…, n, $j \neq i$), the multi-machine system would degenerate into a single-machine system, then the equivalent power deviation of the i-th VSC can be expressed as (6).

$$\begin{aligned} P_{vi}^* - P_{vi} &= I_i^*[Z_{bi}\sin(\theta_{Zbi} + \theta_{Ii}) + Z_p\sin(\theta_{Zp} + \theta_{Ii})] - U_g\sin\delta_i \\ &= U_{tqii} - U_g\sin\delta_i \end{aligned} \tag{6}$$

(6) can be regarded as a nonlinear equation with δ_i as the variable. As mentioned in the existing studies [11], if there is the corresponding δ_i, making P_{vi}^*-$P_{vi} = 0$ hold, it can be proved that the single-VSC system possesses equilibrium points.

Similarly, for a multi-paralleled system, if there are corresponding δ_1, δ_1, …, δ_n, making the system of nonlinear equations (7) hold, then all VSCs in system exist stable equilibrium points.

$$\begin{cases} P_{v1}^* - P_{v1} = -U_g\sin\delta_1 + U_{tq11} + \sum_{j=2}^{n} U_{tqj1} = 0 \\ P_{v2}^* - P_{v2} = -U_g\sin\delta_2 + U_{tq22} + \sum_{j=1,j\neq 2}^{n} U_{tqj2} = 0 \\ \vdots \\ P_{vn}^* - P_{vn} = -U_g\sin\delta_n + U_{tqnn} + \sum_{j=1}^{n-1} U_{tqjn} = 0 \end{cases} \tag{7}$$

The specific calculation method of δ_1, δ_2, …, δ_n can refer to the Newton-Raphson method proposed in literature [14].

3.2. The impact of coupling on equilibrium points

According to (6), when only the i-th VSC is connected to the grid ($\boldsymbol{I}_j^* = 0, j = 1,2\ldots, n, j \neq i$), to satisfy the existence of equilibrium points, P_{vi}^* cannot exceed the maximum or minimum value of P_{vi} ($\min\{P_{vi}\} \leq P_{vi}^* \leq \max\{P_{vi}\}$), otherwise $P_{vi}^* = P_{vi}$ cannot be met. Since P_{vi} reaches its maximum value and minimum value respectively when δ_i is 90° and −90°,the existence range of its equilibrium points can be expressed as (8).

$$-U_g \leqslant U_{tqii} \leqslant U_g \tag{8}$$

where $U_g = \max\{P_{vi}\}$, $-U_g = \min\{P_{vi}\}$.

If U_{tqii} exceeds the range shown in (8), P_{vi}^*-$P_{vi} = 0$ cannot be satisfied, so there is no equilibrium point for the single-VSC system and the synchronization instability incident would occur.

However, for the i-th VSC in a multi-paralleled system, affected by the coupling-synchronization term, the existence range of equilibrium points cannot be simply represented by (8). Because the power flow coupling effects between convertors would influence the existence of its equilibrium point.

By using the sum-difference angle formula and auxiliary angle formula of the trigonometric function, the equivalent power deviation P_{vi}^*-P_{vi} can be rewritten as (9) (see Appendix for more details).

$$P_{vi}^* - P_{vi} = -U_{ei}\sin(\delta_i + \varphi_i) + U_{tqii} \tag{9}$$

And the expression of U_{ei} and φ_i is given by (10).

$$\begin{cases} U_{ei} = \sqrt{a^2 + b^2}, \tan\varphi_i = \dfrac{b}{a} \\ a = U_g + Z_p\sum_{j=1,j\neq i}^{n} I_j^*\cos(\theta_{Ij} + \theta_{Zp} + \delta_j) \\ b = -Z_p\sum_{j=1,j\neq i}^{n} I_j^*\sin(\theta_{Ij} + \theta_{Zp} + \delta_j) \end{cases} \tag{10}$$

Consequently, under the impact of power flow coupling effects, the existence range of equilibrium points for the i-th VSC becomes (11),

$$U_{ei} \leqslant U_{tqii} \leqslant U_{ei} \tag{11}$$

where U_{ei} can be regarded as the stability boundary of the i-th VSC when considering the coupling term. Combined with (10), the specific expression of U_i can be written as (12).

$$U_{ei} = \sqrt{\sum_{j=1,j\neq i}^{n}\left[U_g + Z_pI_j^*\cos(\theta_{Ij} + \theta_{Zp} + \delta_j\right]^2 + \left[Z_p\sum_{j=1,j\neq i}^{n} I_j^*\sin(\theta_{Ij} + \theta_{Zp} + \delta_j)\right]^2} \tag{12}$$

If $U_{ei} = U_g$ is satisfied, which means that the influence of the coupling-synchronization term on the stability boundary of the i-th VSC is minimized, but this circumstance is usually difficult to achieve in practice. If $U_{ei} > U_g$, it indicates that the coupling effect between VSCs would increase the existence range of equilibrium points for the i-th VSC and improve its synchronization stability. On the contrary, if $U_{ei} < U_g$, the coupling effect would reduce the existence range of equilibrium points and deteriorate the synchronization stability of the i-th VSC during LVRT.

3.3. Transient synchronization behavior

To reach the synchronization stability of the whole multi-VSCs system, in addition to satisfying the existence range of equilibrium points, each VSC in the system must also have good enough transient characteristics. For the single-VSC system, existing studies [12] have mentioned that, based on its equivalent rotor swing equation, an equivalent equal area criterion (EAC) can be constructed as a quantitative evaluation method for transient synchronization stability of VSC.

According to (5), due to the similarity of the mathematical model with SG, the equivalent EAC can also be used to analyze the transient synchronization behavior of the VSC in a multi-paralleled system.

Fig. 4 and Fig. 5 show the P_v-δ curve of the i-th VSC under symmetric grid faults, where A_i, S_i, and U_i are the initial operating point, stable equilibrium point, and unstable equilibrium point in the fault-duration stage [9], and δ_{i0}, δ_{is} and δ_{iu} are corresponding EPA, respectively.

When the symmetric fault occurs, the VSC would quickly switch to the current source control model, and follow the grid code to inject negative reactive current ($I_{qi} < 0$) into the grid [10]. Moreover, different

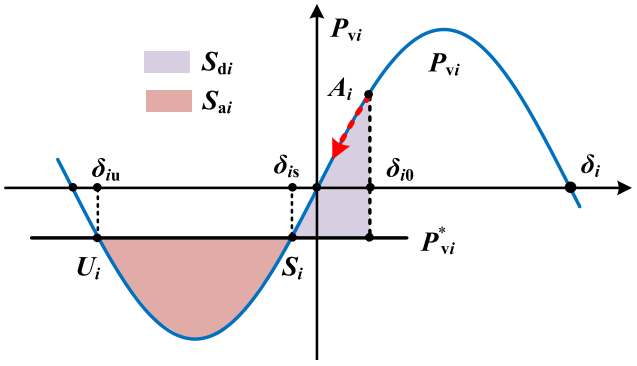


Fig. 4. The P_v-δ curve of the i-th VSC when $\delta_{is} < \delta_{i0}$.

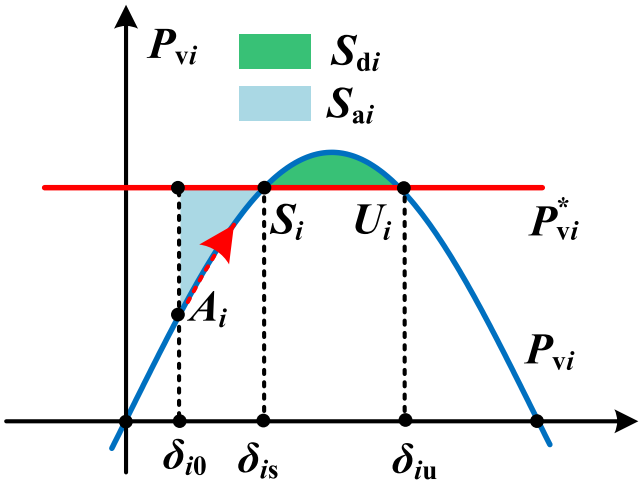


Fig. 5. The P_v-δ curve of the i-th VSC when $\delta_{is} > \delta_{i0}$.

current reference value would directly influence the size of P^*_{vi}, and then change the position of stable equilibrium point S_i (the intersection of P^*_{vi} and P_{vi}), thus affecting the synchronization process of the i-th VSC. There are two cases to be discussed here.

Case A: If S_i is located below A_i ($\delta_{is} < \delta_{i0}$), as shown in Fig. 4, since $P^*_{vi} < P_{vi}$ at point A_i, the insufficiency of equivalent mechanical power would make the i-th VSC undergo a transient synchronization process of deceleration first and then acceleration. In addition, S_{di} and S_{ai} can be regarded as the equivalent deceleration area and equivalent acceleration area.

Case B: Conversely, as shown in Fig. 5, if S_i is located above A_i ($\delta_{is} > \delta_{i0}$), the excess P^*_{vi} would drive the i-th VSC to experience a transient synchronization process of first acceleration and then deceleration.

It can be seen that although the transient synchronization behavior of the above two cases are completely different, the synchronization process has symmetry. Therefore, select Case A as an example for analysis. Hence, the transient synchronization stability criterion of the i-th VSC can be given by (13).

$$\Delta S_i = \underbrace{\int_{\delta_{iu}}^{\delta_{is}} (P^*_{vi} - P_{vi})d\delta_i}_{\max(S_{ai})} + \underbrace{\int_{\delta_{iu}}^{\delta_{i0}} D_i(\omega_{PLLi} - \omega_g)d\delta_i}_{Damping\ \ energy} - \underbrace{\int_{\delta_{is}}^{\delta_{i0}} (P_{vi} - P^*_{vi})d\delta_i}_{S_{di}} \tag{13}$$

ΔS_i can be defined as the stability margin evaluation index. If $\Delta S_i > 0$, it indicates that the i-th VSC has enough potential energy and damping energy to offset the increment in kinetic energy, which can transition to the stable equilibrium point. And the larger ΔS_i is, the greater the stability margin is, and the better the transient synchronization stability of the i-th VSC is. Conversely, if $\Delta S_i < 0$, it means that the equivalent potential energy and damping energy is insufficient to balance the variation of the equivalent kinetic energy for the i-th VSC, so it cannot transition to the stable equilibrium point and would lose synchronization with the grid.

It is worth mentioning that when the i-th VSC is connected to the grid alone, only $\Delta S_i > 0$ is required to ensure the transient stability of the system. But for the multi-paralleled system, only when (14) is satisfied, that is, the transient stability margin of all VSC is greater than 0, the whole system can achieve resynchronization.

$$(\Delta S_1 > 0) \cup (\Delta S_2 > 0) \ldots \cup (\Delta S_n > 0) \tag{14}$$

3.4. The impact of coupling on transient synchronization behavior

According to (5), it can be seen that the positive–negative characteristics of the coupling term will directly affect the size of P^*_{vi}, thereby changing the position of the equilibrium points, and finally affecting the transient synchronization behavior of the i-th VSC. While the positive–negative characteristics of the coupling term is closely related to the current output characteristics of other VSCs, so it is necessary to reveal the change laws of the positive–negative characteristics for the coupling term under different current output conditions. To make the analysis process more intuitive, a two-VSC system shown in Fig. 6 was selected.

Since the work focuses on the current characteristics, four different current output scenarios are considered under the assumption that the control parameters (k_p, k_i) and initial operating status (δ_0) of each VSC are the same. And the following analysis is carried out under the circumstance of $\delta_{is} < \delta_{i0}$.

***Scenario* 1**: $I^*_1 \neq 0$, $I^*_2 = 0 \& \theta_{I1} = -\theta_{Zb1}$.

When θ_{I1} is equal to $-\theta_{Zb1}$, and VSC_2 has no current output, the coupling term would be equal to 0 ($U_{tq21} = 0$). Then, the expression of P^*_{v1} can be expressed as (15).

$$P^*_{v1} = I^*_1 Z_p \sin(\theta_{Zp} + \theta_{I1}) \tag{15}$$

Since the fault degree and line impedance are external factors and cannot be controlled, according to (13), whether the VSC_1 can transition to a stable equilibrium point only depends on its own current output characteristics.

***Scenario* 2**: $I^*_1 = I^*_2 \& \theta_{I1} = -\theta_{Zb1}$, $\theta_{I2} = -\theta_{Zb2}$.

Set current amplitude $I^*_1 = I^*_2$, current phase angle $\theta_{I1} = -\theta_{Zb1}$, $\theta_{I2} = -\theta_{Zb2}$. Then according to the aforementioned assumptions and the iterative calculation result of (7), $\delta_{10} = \delta_{20}$ and $\delta_{1s} = \delta_{2s}$ can be obtained. That is to say, VSC_1 and VSC_2 would be located on the same potential energy surface and would have the same equivalent kinetic energy increment. Therefore, during the transient synchronization process, $\omega_{PLL1} = \omega_{PLL2}$, and further make $\delta_1 = \delta_2$. So, the expression of P^*_{v1} becomes (16),

$$P^*_{v1} = I^*_1 Z_p \sin(\theta_{Zp} + \theta_{I1}) + U_{tq21} \tag{16}$$

where $U_{tq21} = I^*_2 Z_p \sin(\theta_{Zp} + \theta_{I2})$.

Due to the large transmission capacity of the public line, its resistivity is usually smaller than that of the branch line, that is, $\theta_{Zp} > \theta_{Zb2} = -\theta_{I2}$. So $U_{tq21} > 0$ would be satisfied, and P^*_{v1} would increase, and further make the stability margin of VSC_1 increase according to (13). Hence, the coupling effect would play a positive role in VSC_1's transient stability in this case.

***Scenario* 3**: $I^*_1 = I^*_2 \& \theta_{I1} = -\theta_{Zb1}$, $\theta_{I2} < -\theta_{Zb2}$.

The current amplitude of VSC_1 and VSC_2 remains unchanged while reducing the active current proportion of VSC_2, making $\theta_{I2} < -\theta_{Zb2}$. Through the calculation of (7), it can be known that $\delta_{1s} > \delta_{2s}$, so the equivalent potential energy difference of VSC_2 would be larger than that of VSC_1. Therefore, during the transient synchronization process, the increment of equivalent kinetic energy for VSC_2 would be larger than that of VSC_1, and ω_{PLL2} would drop faster than ω_{PLL1}, making δ_2 decrease faster than δ_1 (i.e., δ_2-$\delta_1 < 0$). If the proportion of reactive current in VSC_2 is too high, this may cause $U_{tq21} < 0$ and further lead P^*_{v1} to decrease. So that the VSC_1's transient stability margin ΔS_1 would reduce under this circumstance. Consequently, excessively reducing the active current proportion of VSC_2 would have a negative effect on the VSC_1's

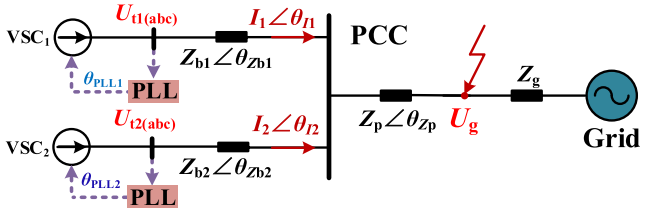


Fig. 6. The structural topology of the two-VSC system.

transient synchronization behavior.

Scenario 4: $I_1^* = I_2^* \& \theta_{I1} = -\theta_{Zb1}$, $\theta_{I2} > -\theta_{Zb2}$.

The current amplitude of VSC_1 and VSC_2 also remains unchanged, but increasing the active current proportion of VSC_2, making $\theta_{I2} > -\theta_{Zb2}$. According to (7), $\delta_{1s} < \delta_{2s}$ can be obtained. So, the equivalent potential energy difference of VSC_2 would be less than that of VSC_1. Hence, the increment of equivalent kinetic energy for VSC_2 would be smaller than that of VSC_1, and ω_{PLL2} would drop slower than ω_{PLL1}, making δ_2 decrease slower than δ_1(i.e., δ_2-$\delta_1 > 0$), causing $U_{tq21} > 0$, and finally leading P_{v1}^* to increase. In this case, the VSC_1's transient stability margin ΔS_1 would also increase. Therefore, reasonably increasing the active current proportion in VSC_2 would contribute to the transient synchronization behavior of VSC_1.

Based on the above analysis, it can be seen that there is a complex power flow coupling effect between VSCs. Different output characteristics would lead to different influence of coupling effect on the VSC transient synchronization stability.

3.5. Stability criterion based on coherence properties

For the multi-paralleled VSCs system, due to the nonlinear dynamic coupling between VSCs, the stability margin of each VSC cannot be calculated directly. That is to say, (13) and (14) can only be used for qualitative analysis, rather than a quantitative evaluation method. To solve this issue, referring to the reduced-order equivalent method of the traditional power system, this article will construct a quantitative stability criterion for the multi-paralleled VSC systems based on coherence characteristics.

In the multi-generators transient stability analysis of traditional power systems [24], it is mentioned that if the power angle difference between any two SGs in a cluster at any time satisfies (17), the cluster is a coherent cluster, and the SGs in the cluster are coherent generators. Usually, ε is 5°~10°. Then, the multi-generators system can be equivalent to a single-machine infinite system.

$$\max\left|\delta_j - \delta_i\right| \leqslant \varepsilon \tag{17}$$

Likewise, if the EPA difference of any two VSCs in the paralleled system meets the (17), it can be reasonably considered that all VSCs in the system are in the coherent status. Then, the coherent cluster can be approximately aggregated into a single VSC system for analysis. Fig. 7 shows the aggregated GFL-VSC.

For this aggregated VSC, its EPA and equivalent output angular frequency can be expressed as

$$\delta_{eq} = \frac{\sum_{i=1}^{n} c_i J_i \delta_i}{\sum_{i=1}^{n} c_i J_i}, \quad \omega_{PLLeq} = \frac{\sum_{i=1}^{n} c_i J_i \omega_{PLLi}}{\sum_{i=1}^{n} c_i J_i} \tag{18}$$

where $c_i = s_i/s_{total}$, s_i is the capacity for i-th VSC and s_{total} is the total capacity for the entire paralleled system.

Moreover, in this coherent cluster, $\delta_1 \approx \delta_2 \approx \ldots \approx \delta_n \approx \delta_{eq}$ and $\omega_{PLL1} \approx \omega_{PLL2} \approx \ldots \approx \omega_{PLLn} \approx \omega_{PLLeq}$ can be obtained. As a result, the equivalent mechanical power and the equivalent output power of the aggregated VSC can be given by (19).

$$\begin{cases} P_{veq}^* = \sum_{i=1}^{n} c_i \{-I_{qi}^* [I_i^* Z_{bi} \sin(\theta_{Zbi} + \theta_{Ii}) + \sum_{j=1}^{n} I_j^* Z_p \sin(\theta_{Zp} + \theta_{Ij})]\} \Big/ \sum_{i=1}^{n} c_i \\ P_{veq} = -I_{qi}^* U_g \sin\delta_{eq} \end{cases} \tag{19}$$

Based on (18) -(19), the dynamic model of the aggregated VSC can be obtained as (20).

$$\begin{cases} J_{eq} \dfrac{d\omega_{PLLeq}}{dt} = P_{veq}^* - P_{veq} - D_{eq}(\omega_{PLLeq} - \omega_g) \\ J_{eq} = \dfrac{\sum_{i=1}^{n} c_i J_i}{\sum_{i=1}^{n} c_i}, D_{eq} = \dfrac{\sum_{i=1}^{n} c_i D_i}{\sum_{i=1}^{n} c_i} \end{cases} \tag{20}$$

According to (20), the P_{veq}-δ_{eq} curve of aggregated VSC can be drawn, as shown in Fig. 8.

It can be seen that (20) still has a similar mathematical form to the rotor motion equation of SG. Therefore, based on the EAC, the transient synchronization stability criterion of the aggregated VSC can be given by (21).

$$\Delta S_{eq} = \underbrace{\int_{\delta_{equ}}^{\delta_{eqs}} (P_{veq}^* - P_{veq}) d\delta_{eq}}_{\max(S_{aeq})} + \underbrace{\int_{\delta_{equ}}^{\delta_{eq0}} D_{eq}(\omega_{PLLeq} - \omega_g) d\delta_i}_{Damping\ energy} - \underbrace{\int_{\delta_{eqs}}^{\delta_{eq0}} (P_{veq} - P_{veq}^*) d\delta_{eq}}_{S_{deq}} \tag{21}$$

When the VSCs in a multi-paralleled system is in coherent status, Criterion (21) can provide quantitative results for the transient synchronization stability assessment of the entire system. If the stability margin $\Delta S_{eq} > 0$ is satisfied, it means that the aggregated VSC can achieve transient synchronization stability during LVRT. Otherwise, due to insufficient stability margin, the entire multi-paralleled VSCs system may face the risk of synchronization instability.

4. Current optimization control strategy for multi-parallel GFL-VSCs system

Based on the above analysis, to realize the synchronization stability of the whole multi-paralleled system during symmetrical gird fault, each VSC must possess a stable equilibrium point and have good transient synchronization behavior at the same time. For this purpose, an improved current optimization control strategy is proposed in this article, which can not only ensure the existence of the equilibrium points but also make the stability margin of each VSC in the paralleled system reaches the maximum, so that the synchronization stability of the whole system reach the optimum. The detailed design idea of the proposed

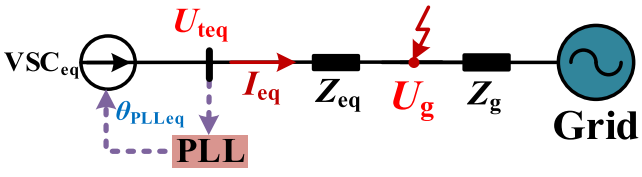


Fig. 7. The aggregated VSC of the multi-paralleled system.

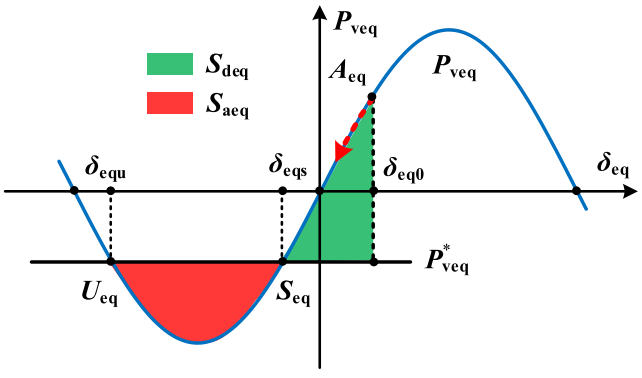


Fig. 8. The P_v-δ curve of the aggregated VSC.

strategy is as follows.

Assuming that each VSC has a stable equilibrium point, and the stable equilibrium point is at the same position as its initial operating point under symmetrical grid fault, as shown in Fig. 9. In this case, since there is no potential energy difference, the equivalent kinetic energy of the each VSC would not change during the fault duration period, ω_{PLL1}, ω_{PLL2}, …, ω_{PLLn} and δ_1, δ_2, …, δ_n remain unchanged. At this point, $S_{d1} = S_{d2}$ …$= S_{dn} = 0$ and S_{a1}, S_{a2}, …, S_{an} get the maximum value, which makes the transient stability margin of each VSC reach the maximum, and leads to the best transient synchronization stability of the entire paralleled system.

According to the above analysis, the derivation process of the proposed current control strategy can be obtained. Combined with (2), substitute $\delta_1 = \delta_{10}$, $\delta_2 = \delta_{20}$, …, $\delta_n = \delta_{n0}$ into (7) and set P^*_{v1}-$P_{v1} = 0$, P^*_{v2}-$P_{v2} = 0$, …, P^*_{vn}-$P_{vn} = 0$. After simplification, (7) can be further expressed as (22),

$$0 = -U + C_{P_n\times n} I^*_{d_n\times 1} + D_{P_n\times n} I^*_{q_n\times 1} \tag{22}$$

where $\boldsymbol{I}^*_{d_n\times 1}$ is the active current matrix, $\boldsymbol{I}^*_{d_n\times 1} = [I^*_{d1}, I^*_{d2},\ldots, I^*_{dn}]^T$;

$\boldsymbol{I}^*_{q_n\times 1}$ is the reactive current matrix, $\boldsymbol{I}^*_{q_n\times 1} = [I^*_{q1}, I^*_{q2},\ldots, I^*_{qn}]^T$; $\boldsymbol{U}$ is the grid voltage matrix, $\boldsymbol{U} = U_g\,[\sin\delta_{10}, \sin\delta_{20}, \ldots, \sin\delta_{n0}]^T$. The $\boldsymbol{C}_{P_n\times n}$ and $\boldsymbol{D}_{P_n\times n}$ matrices are as follows.

$$C_{P_n\times n} = \begin{bmatrix} Z_1\sin\theta_{Zb1}+Z\sin\theta_Z & Z\sin(\theta_Z+\delta_{20}-\delta_{10}) & \cdots & Z\sin(\theta_Z+\delta_{n0}-\delta_{10}) \\ Z\sin(\theta_Z+\delta_{10}-\delta_{20}) & Z_2\sin\theta_{Zb2}+Z\sin\theta_Z & \cdots & Z\sin(\theta_Z+\delta_{n0}-\delta_{20}) \\ \vdots & \vdots & \vdots & \vdots \\ Z\sin(\theta_Z+\delta_{10}-\delta_{n0}) & Z\sin(\theta_Z+\delta_{20}-\delta_{n0}) & \cdots & Z_n\sin\theta_{Zbn}+Z\sin\theta_Z \end{bmatrix} \tag{23}$$

$$D_{P_n\times n} = \begin{bmatrix} Z_1\cos\theta_{Zb1}+Z\cos\theta_Z & Z\cos(\theta_Z+\delta_{20}-\delta_{10}) & \cdots & Z\cos(\theta_Z+\delta_{n0}-\delta_{10}) \\ Z\cos(\theta_Z+\delta_{10}-\delta_{20}) & Z_2\cos\theta_{Zb2}+Z\cos\theta_Z & \cdots & Z\cos(\theta_Z+\delta_{n0}-\delta_{20}) \\ \vdots & \vdots & \vdots & \vdots \\ Z\cos(\theta_Z+\delta_{10}-\delta_{n0}) & Z\cos(\theta_Z+\delta_{20}-\delta_{n0}) & \cdots & Z_n\cos\theta_{Zbn}+Z\cos\theta_Z \end{bmatrix} \tag{24}$$

Since the information of line impedance, grid voltage, and initial EPA can be measured [25], which are all known quantities, and the reactive current reference value of each VSC is determined by the grid code [22], as shown in (25).

$$I^*_{qi} = -1.5\times(0.9-U_g), \quad \text{if } \left|I^*_{qi}\right| > 1.0 \text{ p.u.}, \quad I^*_{qi} = -1.0 \text{ p.u.} \tag{25}$$

Therefore, according to (22) - (25), the active current reference of each VSC required to meet the maximum stability margin can be given by (26).

$$I^*_{d_n\times 1} = -C^{-1}_{P_n\times n} D_{P_n\times n} I^*_{q_n\times 1} + C^{-1}_{P_n\times n} U \tag{26}$$

Furthermore, it is worth noting that if the current reference value calculated according to (25) and (26) exceeds the converter capacity limit shown in (27), priority should be given to meeting the reactive current requirements of the grid code, and the active current reference value should be reduced to within the remaining capacity of the converter,

$$\sqrt{\left(I^*_{di}\right)^2 + \left(I^*_{qi}\right)^2} < I_M \tag{27}$$

where I_M is the maximum current value allowed for the operation of the VSC, which is generally 1.1 times the rated current [14].

Fig. 10 shows the control block diagram of the proposed current optimization method. In the pre-fault stage, the fault flag is 0, the VSCs are at normal operating status. When a severe symmetric grid fault occurs, the fault flag would be switched to 1, and the proposed strategy would be adopted. Moreover, the current flag is used to ensure that the current reference value of VSC does not exceed the current limitation.

It should be noted that the proposed strategy also has some limitations. Firstly, the current reference amplitude calculated according to (25) and (26) cannot exceed the current limit of the converter. Secondly, this current optimization strategy is based on the radial circuit topology, and how to apply it to circular circuit topology needs further study. Finally, the proposed strategy is only applicable to multi-paralleled GFL-VSCs system. If the grid-forming VSC (GFM-VSC) exists in the system, owing to the differences in external characteristics, the proposed strategy cannot be used.

5. Simulation validation

To validate the analytical results and the effectiveness of the proposed strategy, the simulations on a three-paralleled VSCs system were performed using MATLAB/Simulink. The simulated system was configured, as shown in Fig. 1, and Table 1. and 2 of the Appendix shows the detailed parameters of the simulation system.

5.1. Fast response characteristics of current loop

At 0.1 s, the grid voltage U_g drops symmetrically to 0.2p.u., taking VSC_1 as an example, the fast response characteristics of current loop is verified by simulation Case 1.

Case 1: I^*_{d1} and I^*_{q1} are set to 0.3p.u. and −1.0p.u., I^*_{d2}, I^*_{q2}, I^*_{d3} and I^*_{q3} are all set to 0p.u. under grid fault. Since VSC_2 and VSC_3 have no current output, the three-VSC paralleled system degenerates into a single-VSC system. With sufficient stability margin [9], VSC_1 resynchronizes with grid during LVRT. Besides, as shown in Fig. 11 (d), the output active/reactive currents quickly reach the reference values

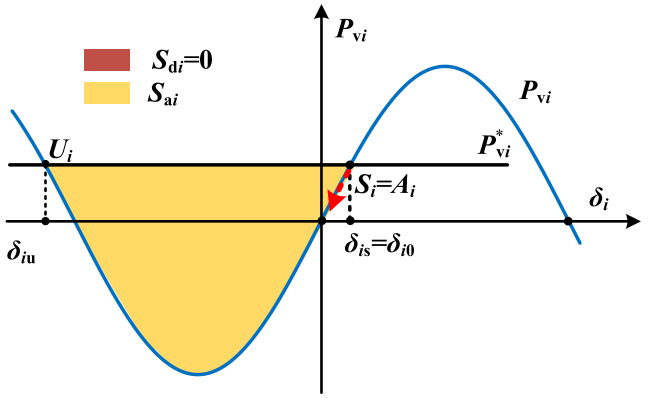


Fig.9. The P_v-δ curve of the i-th VSC when $\delta_{is} = \delta_{i0}$.

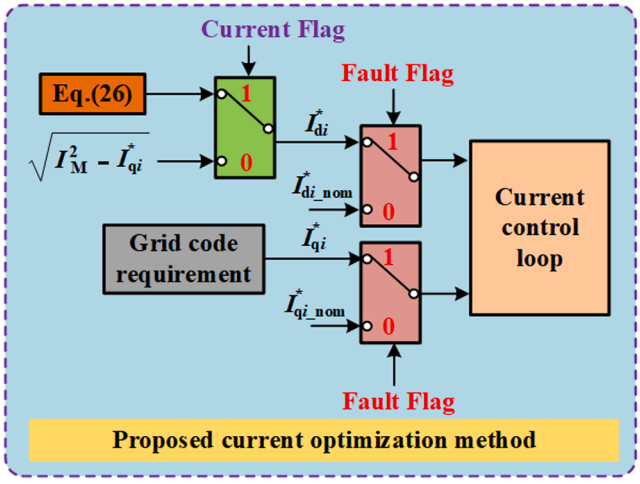


Fig. 10. Control block diagram of the proposed current optimization method.

Table 1
The parameters of the simulation system.

Symbol	Description	Value (p.u.)
P_n	Rate power	2 MW (1p.u.)
V_n	Rate voltage	690 V (1p.u.)
V_{dc}	dc-link voltage	1200 V
f_g	Rate frequency	50 Hz
L_f	Filter inductance	0.25p.u.
C_f	Filter capacitor	0.15p.u.
k_p, k_i	PI coefficients of the PLL	1.0, 100
R_{b1}	Brunch transmission line resistance	0.1968p.u.
X_{b1}	Brunch transmission line reactance	0.5543p.u.
R_{b2}	Brunch transmission line resistance	0.1980p.u.
X_{b2}	Brunch transmission line reactance	0.5539p.u.
R_{b3}	Brunch transmission line resistance	0.1868p.u.
X_{b3}	Brunch transmission line reactance	0.5222p.u.
R_P	Public transmission line resistance	0.0118p.u.
X_P	Public transmission line reactance	0.0426p.u.
Z_{base}	Impedance base value	0.23805 Ω

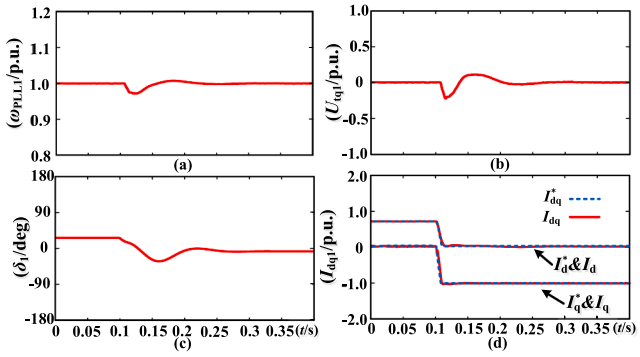


Fig. 11. Simulation results of Case 1. (a) PLL's output angular frequency of VSC_1. (b) The q-axis component of VSC_1's terminal voltage. (c) The EPA of VSC_1. (d) The dq-axis current of VSC_1.

after a very short regulating process, much faster than the variation rate of ω_{PLL} shown is Fig. 11 (a). Thus, the inner current controllers could be considered ideal in the transient synchronization stability analysis with the PLL as the key factor.

5.2. Simulation validation of synchronization stability and coupling characteristic analysis

The influence of power flow coupling on the VSC's equilibrium point is verified by simulation Case 2-Case 5.

Case 2: After the grid fault occurs, I_{d1}^* and I_{q1}^* are set to 0p.u. and −1.0p.u., I_{d2}^*, I_{q2}^*, I_{d3}^* and I_{q3}^* are all set to 0p.u. By calculation, U_{tq11} = -0.2086p.u., less than $-U_g$, so (8) can't be satisfied. It means that the equilibrium point of VSC_1 is missing and the synchronization instability caused by the lack of equilibrium points would occur. The simulation results shown in Fig. 12 also verify the above analysis. Since the operating point exceeds the synchronization stability boundary, VSC_1 suffers from an instability phenomenon for angular frequency drop.

Case 3: I_{d1}^* and I_{q1}^* are set to 0p.u. and −1.0p.u., I_{d2}^* and I_{q2}^* are set to 0.3p.u. and −1.0p.u., I_{d3}^* and I_{q3}^* are set to 0.3p.u. and −1.0p.u., respectively. Based on (7) and (12), $-U_{e1}$ = -0.2975p.u., while U_{tq11} = -0.2086p.u., meeting the range shown in (11). Compared with Case 2, it can be seen that the stability boundary of VSC_1 is increased, which means that the coupling effect improves the synchronization stability of VSC. Fig. 12 shows the simulation results, under the impact of the coupling effect, the VSC_1 avoids the synchronization instability incident.

Case 4: I_{d1}^* and I_{q1}^* are set to 0.16p.u. and −1.0p.u., I_{d2}^*, I_{q2}^*, I_{d3}^* and I_{q3}^* are all set to 0p.u. The paralleled system becomes a single-

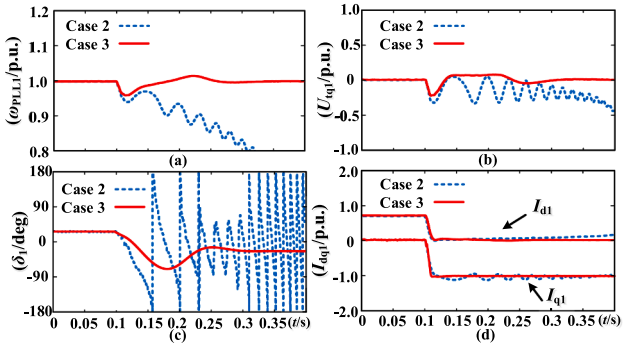


Fig. 12. Simulation results of Case 2 and Case 3. (a) PLL's output angular frequency of VSC_1. (b) The q-axis component of VSC_1's terminal voltage. (c) The EPA of VSC_1. (d) The dq-axis current of VSC_1.

VSC system as well. At this point, U_{tq11} = -0.1154p.u., larger than $-U_g$, satisfying (8). In addition, according to (13), the transient stability margin of VSC_1 is also>0. So, the VSC_1 can achieve synchronization stability under grid fault. Fig. 13 shows the simulation results. It can be seen that VSC_1 has successfully transitioned to the new stable equilibrium point.

Case 5: I_{d1}^* and I_{q1}^* are set to 0.16p.u. and −1.0p.u., I_{d2}^* and I_{q2}^* are set to 0p.u. and −1.0p.u., I_{d3}^* and I_{q3}^* are set to 0p.u. and −1.0p.u., respectively. Through calculation, (7) has no solution, which means that there is no equilibrium point for VSC_1 at this time, so that the instability incident would occur. The simulation results shown in Fig. 13 also validates the above analysis. Affected by the coupling effect, VSC_1 loses its equilibrium points, resulting in synchronization instability incident.

Secondly, the influence of power flow coupling on transient synchronization behavior is verified by simulation Case 6-Case 9. Before the symmetrical fault occurs, $\delta_{10} \approx \delta_{20} \approx \delta_{30}$.

Case 6: I_{d1}^* and I_{q1}^* are set to 0.3346p.u. and −0.9424p.u. ($\theta_{I1} = -\theta_{Zb1}$), I_{d2}^*, I_{q2}^*, I_{d3}^* and I_{q3}^* are all set to 0p.u., respectively. Since the system degenerates into a single-machine system, the transient stability of VSC_1 depends on its own current output characteristics. According to (13), ΔS_1 is>0, so that VSC_1 would avoid transient instability incident. Fig. 14 shows the simulation results, due to VSC_1 has enough potential energy to offset the increment of kinetic energy, it resynchronizes with the grid during LVRT.

Case 7: I_{d1}^* and I_{q1}^* are set to 0.3346p.u. and −0.9424p.u.,

I_{d2}^* and I_{q2}^* are set to 0.3366p.u. and −0.9416p.u. ($\theta_{I2} = -\theta_{Zb2}$),

I_{d3}^* and I_{q3}^* are set to 0.3368p.u. and −0.9415p.u. ($\theta_{I3} = -\theta_{Zb3}$), respectively, and $I_1^* = I_2^* = I_3^* = 1.0$p.u.. Combined with the analysis in Scenario 2 (Section 3.4), VSC_1, VSC_2 and VSC_3 would have the same transient response process under such condition, so according to (13),

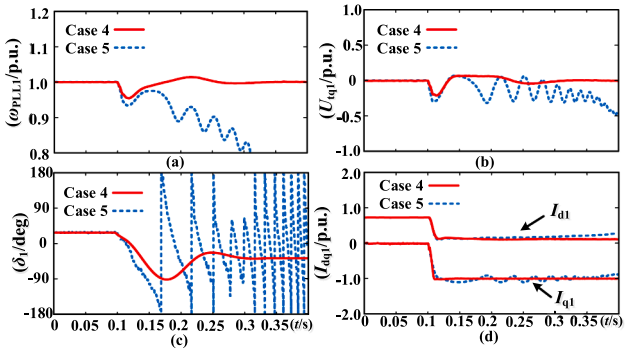


Fig. 13. Simulation results of Case 4 and Case 5. (a) PLL's output angular frequency of VSC_1. (b) The q-axis component of VSC_1's terminal voltage. (c) The EPA of VSC_1. (d) The dq-axis current of VSC_1.

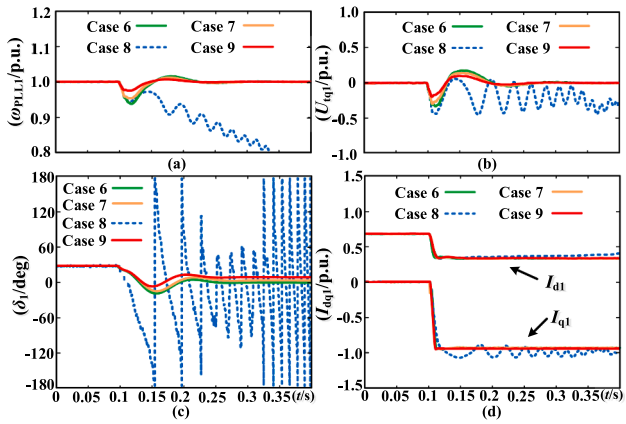


Fig. 14. Simulation results for Case 6 - Case 9. (a) PLL's output angular frequency of VSC_1. (b) The q-axis component of VSC_1's terminal voltage. (c) The EPA of VSC_1. (d) The dq-axis current of VSC_1.

ΔS_1 is slightly increased. Therefore, in this case, the coupling effect would increase the transient stability margin of the VSC_1 lightly. Fig. 14 shows the simulation results, the transient synchronization stability of VSC_1 is slightly enhanced.

Case 8: I^*_{d1} and I^*_{q1} are set to 0.3346p.u. and −0.9424p.u., I^*_{d2} and I^*_{q2} are set to 0p.u. and −1.0p.u. ($\theta_{I2} < -\theta_{Zb2}$), I^*_{d3} and I^*_{q3} are set to 0p.u. and −1.0p.u. ($\theta_{I3} < -\theta_{Zb3}$), respectively. Based on the analysis in Scenario 3 (Section 3.4), excessive reducing the active current ratio of VSC_2 and VSC_3 would decrease the stability margin of VSC_1 and deteriorate its transient synchronization stability. Fig. 14 shows the simulation results, due to the negative impact of the coupling effects, VSC_1 gradually loses synchronization with the grid during LVRT.

Case 9: I^*_{d1} and I^*_{q1} are set to 0.3346p.u. and −0.9424p.u., I^*_{d2} and I^*_{q2} are set to 0.4359p.u. and −0.90p.u. ($\theta_{I2} > -\theta_{Zb2}$), I^*_{d3} and I^*_{q3} are set to 0.4359p.u. and −0.90p.u. ($\theta_{I3} > -\theta_{Zb3}$), respectively, also $I^*_1 = I^*_2 = I^*_3 =$ 1.0 p.u. According to the analysis in Scenario 4 (Section 3.4), properly increasing the proportion of active current in VSC_2 and VSC_3 would increase the transient stability margin of VSC_1 and improve its synchronization stability. Fig. 14 shows the simulation results, in this case, the coupling effect significantly enhances the resynchronization ability of VSC_1, making its transient transition process smoother.

Subsequently, the stability criterion based on coherence properties for the multi-paralleled VSCs system is validated by Case 10 and Case 11.

Case 10: I^*_{d1}, I^*_{d2} and I^*_{d3} are all set to 0.3p.u., I^*_{q1}, I^*_{q2} and I^*_{q3} are all set to −1.0p.u. At this point, (7) has solutions, where δ_{1s} = -7.99°, δ_{2s} = -8.21°, and δ_{3S} = -7.76°, that is, each VSC has a stable equilibrium point. Since each VSC has the same initial status, the similar position of stable equilibrium point, and the consistent control parameters, it can be reasonably considered that VSC in the paralleled system has similar transient response process, satisfying the coherence principle shown in (17). Thus, criterion (21) can be used for transient stability assessment of the entire system. According to, ΔS_{eq} is greater than 0, which means that the whole system can achieve transient stability during the fault. Fig. 15 shows the simulation results. It can be seen that the paralleled system resynchronizes with the grid, and the transient synchronization process of each VSC is almost identical.

Case 11: I^*_{d1}, I^*_{d2} and I^*_{d3} are all set to 0.1p.u., I^*_{q1}, I^*_{q2} and I^*_{q3} are all set to −1.0p.u.. (7) still has solutions, where δ_{1s} = -56.36°, δ_{2s} = -56.65° and δ_{3S} = -53.78°. Similar to the analysis in Case 10, the paralleled system satisfies the coherence principle. Then, according to (21), ΔS_{eq}<0. This means that the entire system would suffer from transient instability during grid fault. Fig. 16 shows the simulation results, each VSC in the system is lose synchronization with the grid and has a similar LOS phenomenon.

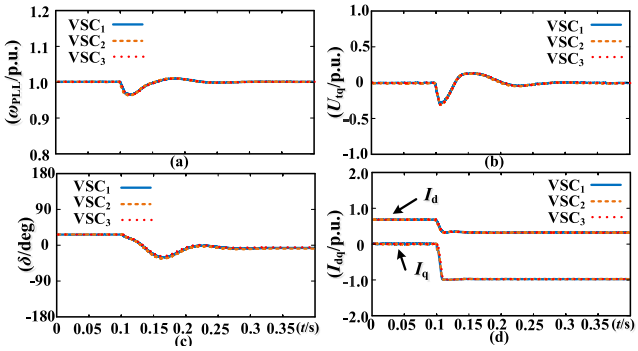


Fig. 15. Simulation results for Case 10. (a) PLL's output angular frequency of VSC_i (i = 1,2,3). (b) The q-axis component of terminal voltage for VSC_i. (c) The EPA of VSC_i. (d) The dq-axis current of VSC_i.

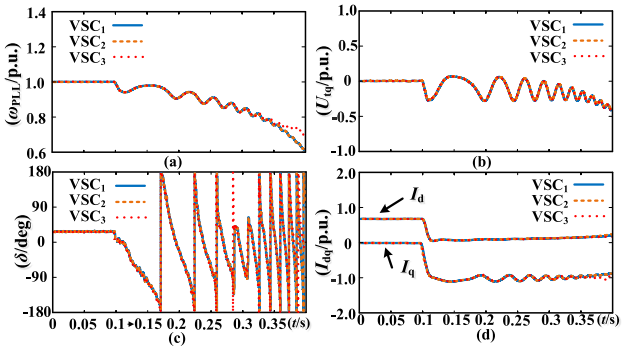


Fig. 16. Simulation results for Case 11. (a) PLL's output angular frequency of VSC_i (i = 1,2,3). (b) The q-axis component of terminal voltage for VSC_i. (c) The EPA of VSC_i. (d) The dq-axis current of VSC_i.

5.3. Simulation validation of the proposed method

To demonstrate the effectiveness of the proposed current optimization method, comparisons with improved PLL based on additional damping [10], adaptive active current control strategy [8], and autonomic current distribution scheme [14] have been performed in this part. To make synchronization instability significant, the grid voltage U_g drops to 0.1p.u.

Improved PLL based on additional damping

Each VSC injects 1.0p.u. pure reactive current into the grid, and according to (7), there is no stable equilibrium point for each VSCs. As shown by the blue dotted line in Fig. 17, even if the additional damping control strategy is adopted, the synchronization instability incident for VSCs cannot be avoided in the absence of the equilibrium point.

Adaptive active current control strategy

This strategy can adaptively adjust VSC's active current reference value based on the change trend of ω_{PLL}, thereby directly improving the output characteristics of VSC, so that the VSC can avoid the risk of LOS caused by the loss of equilibrium point and poor transient synchronization behavior. As shown by the solid green line in Fig. 17, the VSCs resynchronize with the power grid, but due to the introduction of the PLL dynamics, the active current fluctuates significantly, resulting in non-ideal dynamic characteristics for the VSCs.

Autonomic current distribution scheme

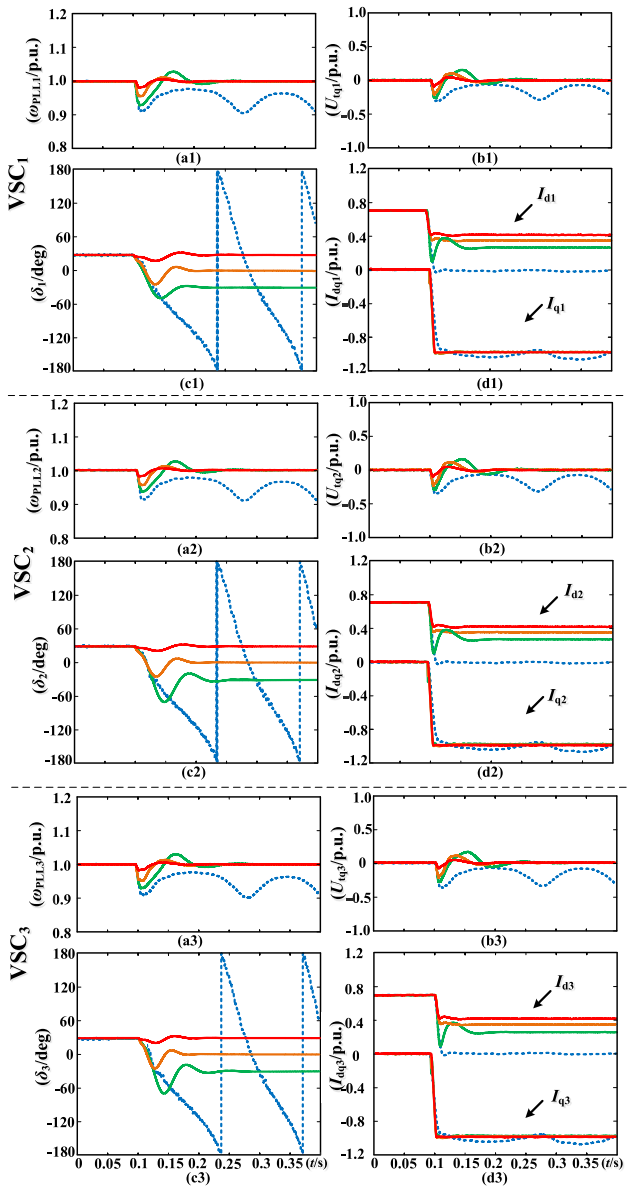


Fig. 17. Simulation results for each with different control method. (a1), (a2), and (a3) PLL's output angular frequency. (b1), (b2), and (b3) The q-axis component of terminal voltage. (c1), (c2), and (c3) The EPA. (d1), (d2), and (d3) The dq-axis current.

Based on the line impedance information and grid code, I_{d1}^{*} and I_{q1}^{*} are set to 0.3393p.u. and −1.0p.u., I_{d2}^{*} and I_{q2}^{*} are set to 0.3418p.u. and −1.0p.u., I_{d3}^{*} and I_{q3}^{*} are set to 0.3405p.u. and −1.0p.u., respectively. As shown by the solid yellow line in Fig. 17, the EPA of each VSC is close to 0°, making the multi-paralleled system avoid LOS. But the ω_{PLL} and δ still have relatively large overshoot and oscillation during the regulation process.

Proposed method

When the proposed method is applied, I_{d1}^{*} and I_{q1}^{*} are set to 0.41p.u. and −1.0p.u., I_{d2}^{*} and I_{q2}^{*} are set to 0.4125p.u. and −1.0p.u., I_{d3}^{*} and I_{q3}^{*} are set to 0.4181p.u. and −1.0p.u., respectively. As shown by the solid red line in Fig. 17, Since the transient stability margin of each VSC reaches the maximum, the overshoot and oscillation of ω_{PLL} and δ are effectively suppressed, making the transient synchronization process of VSCs quite fast and smooth. Consequently, the proposed strategy can effectively improve the dynamic performance of VSCs and significantly enhance the transient synchronization stability of the entire system.

6. Conclusion

This study investigated the power flow coupling mechanism and transient synchronization stability optimization control method of the multi-paralleled GFL-VSCs. The major findings are as follows.

1) The existence of equilibrium points and transient synchronization behavior of converters is not only closely related to its own current output characteristics during LVRT, but also affected by the power flow coupling effect between VSCs. By considering different current output scenarios, this article reveals the influence of power flow coupling effect on the VSC's transient synchronization stability.
2) When all VSCs are in a coherent status, the multi-paralleled system can be equivalent to a single VSC system through reduced-order aggregation. Therefore, a quantitative assessment method can be constructed, which can evaluate the transient synchronization stability of a multi-paralleled system.
3) In order to maximize the transient synchronization stability margin of each VSC, a current optimization control strategy is proposed in this article. By adjusting the current reference value of VSCs reasonably with the proposed method, the transient synchronization stability of the entire paralleled system can be ensured to be the best. The comparison with the existing typical strategies also verifies that the proposed strategy has pretty good control performance.

In addition to GFL-VSC, the transient stability analysis and control of multi-paralleled GFM-VSCs system is also worthy of attention. However, due to the high-order model and complex coupling of multi-converters system, how to establish accurate and easy-to-use stability assessment criteria is still facing challenges, so this topic is also a major focus of the future work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix

The q-axis component of terminal voltage U_{tqi} can be expressed as,

$$\begin{aligned} U_{tqi} &= -U_g\sin\delta_i + \underbrace{I_i^*[Z_{bi}\sin(\theta_{Zbi}+\theta_{Ii}) + Z_p\sin(\theta_{Zp}+\theta_{Ii})]}_{U_{tqii}} \\ &\quad + \sum_{j=1,j\neq i}^{n} \underbrace{Z_p I_j^*\sin(\theta_{Zp}+\theta_{Ij}+\delta_j-\delta_i)}_{U_{tqji}} \end{aligned} \tag{A1}$$

Using the sum difference angle formula of trigonometric functions, $Z_pI_j^*\sin(\theta_{Zp}+\theta_{Ij}+\delta_i\text{-}\delta_j)$ can be rewritten as:

$$U_{tqji} = Z_pI_j^*[\sin(\theta_{Zp}+\theta_{Ij}+\delta_j)\cos\delta_i - \cos(\theta_{Zp}+\theta_{Ij}+\delta_j)\sin\delta_i] \tag{A2}$$

Substituting (A2) into (A1),(A3) can be obtained.

$$\begin{aligned} U_{tqi} &= -U_g\sin\delta_i + U_{tqii} \\ &\quad + \sum_{j=1,j\neq i}^{n} Z_pI_j^*[\sin(\theta_{Zp}+\theta_{Ij}+\delta_j)\cos\delta_i - \cos(\theta_{Zp}+\theta_{Ij}+\delta_j)\sin\delta_i] \end{aligned} \tag{A3}$$

By merging the items containing $\cos\delta_i$ and $\sin\delta_i$ respectively, (A3) can be rewritten as (A4).

$$\begin{aligned} U_{tqi} &= U_{tqii} - \Bigg\{ \underbrace{\left[-Z_p\sum\nolimits_{j=1,j\neq i}^{n} I_j^*\sin\left(\theta_{Ij}+\theta_{Zp}+\delta_j\right)\right]}_{b}\cos\delta_i \\ &\quad + \underbrace{\left[U_g + Z_p\sum\nolimits_{j=1,j\neq i}^{n} I_j^*\cos\left(\theta_{Ij}+\theta_{Zp}+\delta_j\right)\right]}_{a}\sin\delta_i \Bigg\} \end{aligned} \tag{A4}$$

The last two terms on the right side of (A4) can be combined using the auxiliary angle formula of trigonometric functions.

$$U_{tqi} = P_{vi}^* - P_{vi} = -U_{ei}\sin(\delta_i+\varphi_i) + U_{tqii} \tag{A5}$$

And the expression of U_{ei} and φ_i is given by (A6).

$$\begin{cases} U_{ei} = \sqrt{a^2+b^2}\,, \tan\varphi_i = \dfrac{b}{a} \\ a = U_g + Z_p\sum\limits_{j=1,j\neq i}^{n} I_j^*\cos(\theta_{Ij}+\theta_{Zp}+\delta_j) \\ b = -Z_p\sum\limits_{j=1,j\neq i}^{n} I_j^*\sin(\theta_{Ij}+\theta_{Zp}+\delta_j) \end{cases} \tag{A6}$$

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